Optimization Formulation Specification

> **EMS Readiness Optimization – Phase 3**
>
> This document describes the mathematical formulations used to determine optimal EMS unit allocation across Manhattan FDNY firehouses.

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1. Notation

Symbol	Description
I	Set of firehouses (staging locations), $\lvert I \rvert = 48$
J	Set of demand precincts, $\lvert J \rvert = 25$
K	Total number of EMS units to allocate
C	Maximum units per firehouse (capacity)
d_j	Demand weight for precinct j (arrival rate λ_j)
t_{ij}	Travel time (minutes) from firehouse i to precinct j
τ	Coverage threshold (minutes) for the maximal-coverage model
a_{ij}	Coverage indicator: 1 if $t_{ij} \le \tau$, else 0

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2. Formulation A – Demand-Weighted Allocation

Goal: Minimise the total demand-weighted expected response time.

Decision Variables

- $x_i \in {0, 1, \dots, C}$ — number of units at firehouse i
- $y_{ij} \in {0, 1}$ — 1 if precinct j is served by firehouse i

Objective

$$\min \sum_{j \in J} \sum_{i \in I} d_j \, t_{ij} \, y_{ij}$$

Constraints

#	Constraint	Meaning
1	$\sum_{i \in I} x_i = K$	Deploy exactly K units
2	$0 \le x_i \le C,\; x_i \in \mathbb{Z}$	Capacity and integrality
3	$\sum_{i \in I} y_{ij} = 1 \; \forall j$	Each precinct served by exactly one firehouse
4	$y_{ij} \le x_i \;\forall i,j$	Can only assign to an open firehouse

Interpretation

This is the primary model for **P2 (optimised fixed staging)**. It simultaneously decides how many units to place at each firehouse and which firehouse serves each precinct, minimising the demand-weighted travel time.

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3. Formulation B – P-Median Model

Goal: Select exactly K firehouses and assign precincts to minimise total demand-weighted distance.

Decision Variables

- $x_i \in \{0, 1\}$ — 1 if firehouse i is opened
- $y_{ij} \in \{0, 1\}$ — 1 if precinct j is served by firehouse i

Objective

$$\min \sum_{j \in J} \sum_{i \in I} d_j \, t_{ij} \, y_{ij}$$

Constraints

#	Constraint	Meaning
1	$\sum_{i \in I} x_i = K$	Open exactly K firehouses
2	$\sum_{i \in I} y_{ij} = 1 \; \forall j$	Each precinct served by exactly one firehouse
3	$y_{ij} \le x_i \;\forall i,j$	Can only assign to an open firehouse

Interpretation

The p-median answers: “If we can only staff K firehouses, which K should we choose?” This is useful when fixed costs dominate and units are scarce. Note that here K refers to the number of **open locations**, not the total unit count.

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4. Formulation C – Maximal Coverage Model

Goal: Maximise the total demand that is covered within a travel-time threshold τ.

Decision Variables

- $x_i \in {0, 1, \dots, C}$ — number of units at firehouse i
- $z_j \in {0, 1}$ — 1 if precinct j is covered

Objective

$$\max \sum_{j \in J} d_j \, z_j$$

Constraints

#	Constraint	Meaning
1	$\sum_{i \in I} x_i = K$	Deploy exactly K units
2	$0 \le x_i \le C,\; x_i \in \mathbb{Z}$	Capacity and integrality
3	$z_j \le \sum_{i \in I} x_i \, a_{ij} \;\forall j$	Precinct covered only if a nearby firehouse has units

Interpretation

The maximal-coverage model focuses on a binary service-level target: “Can we reach precinct j in $\le \tau$ minutes?” With $\tau = 8$ minutes (configurable), it maximises the proportion of demand that meets this standard. This model is especially useful for equity analysis and minimum service-level guarantees.

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5. Baseline Policies (Non-Optimisation)

P0 – Uniform Allocation

$$x_i = \lfloor K / |I| \rfloor$$

Remainder units distributed round-robin. Ignores demand patterns; serves as a naïve lower bound.

P1 – Demand-Proportional Allocation

$$x_i \propto \sum_{j : i = \text{nearest}(j)} d_j$$

Each firehouse receives units proportional to the demand of precincts for which it is the closest firehouse. Rounded to integers respecting capacity.

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6. Solver Details

Property	Value
Library	PuLP 3.x (Python)
Default solver	CBC (COIN-OR Branch-and-Cut)
Fallback solver	GLPK
Time limit	300 seconds
Optimality gap	Default (solver-determined)

CBC is bundled with PuLP and requires no external installation. Problem sizes (48×25 variables) are well within CBC's capabilities; typical solve times are under 5 seconds.

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7. Implementation Map

Formulation	Source file	Function
Demand-Weighted	`optimization/models.py`	`build_demand_weighted()`
P-Median	`optimization/models.py`	`build_p_median()`
Maximal Coverage	`optimization/models.py`	`build_maximal_coverage()`
Uniform baseline	`optimization/policies.py`	`uniform_allocation()`
Demand-proportional	`optimization/policies.py`	`demand_proportional_allocation()`
High-level runner	`optimization/allocator.py`	`EMSAllocator.solve()`
Configuration	`configs/optimization.yaml`	—

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8. Use Cases

Policy	Model	When to use
P0	Uniform	Naïve baseline / equity benchmark
P1	Demand-proportional	Quick heuristic; no solver needed
P2	Demand-weighted	**Primary recommendation** – optimal fixed staging
P2-alt	P-median	Selecting K locations from 48 candidates
P2-alt	Maximal coverage	Service-level / equity analysis
P3	Time-varying	Future phase (hour-specific $ \lambda$ tables)

References

1. Daskin, M.S. (2013). Network and Discrete Location. Wiley.
2. Church, R. & ReVelle, C. (1974). The Maximal Covering Location Problem. Papers of the Regional Science Association, 32, 101–118.
3. ReVelle, C. & Swain, R. (1970). Central facilities location. Geographical Analysis, 2(1), 30–42.